Overview of Entropic Neural OT via Diffusion Processes

Let $\mathcal{X}$ and $\mathcal{Y}$ be two d-dimensional Euclidean spaces, and let $\mathcal{P}_2(\mathcal{X})$ and $\mathcal{P}_2(\mathcal{Y})$ denote their respective spaces of Borel probability measures with finite second moments. Consider $P_0 \in \mathcal{P}_2(\mathcal{X})$ and $P_1 \in \mathcal{P}_2(\mathcal{Y})$ as two prescribed marginal distributions.

Let $\Omega = C([0,1], \mathbb{R}^d)$ be the path space of continuous functions describing trajectories in $\mathbb{R}^d$ from time $t = 0$ to $t = 1$. Let $\mathcal{P}(\Omega)$ denote the set of all probability measures on Ω.

Let $W^\epsilon \in \mathcal{P}(\Omega)$ be our reference measure, defined as the law of a scaled Wiener process with diffusion coefficient $\sqrt{\epsilon} > 0$ initialized at P_0. Specifically, W^ϵ is the law of the continuous-time process $(X_t)_{t\in[0,1]}$ governed by the following stochastic differential equation (SDE):

$$\mathrm{d}X_t = \sqrt{\epsilon}\mathrm{d}W_t, \quad X_0 \sim P_0$$

where $(W_t)_{t\in[0,1]}$ is a standard d-dimensional Brownian motion.

Let $T \in \mathcal{P}(\Omega)$ be the law of a stochastic process. Let $\pi^T \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ denote its joint distribution at the endpoints $t = 0$ and $t = 1$, and let $\pi_0^T \in \mathcal{P}(\mathcal{X})$ and $\pi_1^T \in \mathcal{P}(\mathcal{Y})$ be its marginal distributions at times $t = 0$ and $t = 1$, respectively. Furthermore, let $T_{|x,y} \in \mathcal{P}(\Omega)$ denote the conditional law of the process T given the endpoint states $X_0 = x$ and $X_1 = y$.

Then, by decomposing the Kullback-Leibler (KL) divergence on the path space (following the authors of Entropic Neural Optimal Transport via Diffusion Processes), it holds that:

$$\mathrm{KL}(T \mid W^\epsilon) = \mathrm{KL}(\pi^T \mid \pi^{W^\epsilon}) + \int_{\mathcal{X}\times\mathcal{Y}} \mathrm{KL}(T_{|xy} \mid W^\epsilon_{|xy})\mathrm{d}\pi^T(x,y)$$

Let $\mathcal{R}(W^\epsilon) \subset \mathcal{P}(\Omega)$ denote the reciprocal class of the reference measure W^ϵ. A probability measure T belongs to $\mathcal{R}(W^\epsilon)$ if it shares the same conditional bridge dynamics as W^ϵ, meaning:

$$T_{|xy} = W^\epsilon_{|xy}, \quad \text{for all} \quad (x,y) \in \mathcal{X} \times \mathcal{Y}$$

Let $\mathcal{F}(P_0, P_1) \subset \mathcal{P}(\Omega)$ be the space of all path measures on Ω having marginals P_0 and P_1 at times $t = 0$ and $t = 1$, respectively. Let $\Pi(P_0, P_1) \subset \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ denote the space of all couplings (joint distributions) on $\mathcal{X} \times \mathcal{Y}$ with marginals P_0 and P_1.

The dynamic Schrödinger Problem seeks to find the process $T \in \mathcal{F}(P_0, P_1)$ that minimizes $\mathrm{KL}(T \mid W^\epsilon)$. Using the KL decomposition from earlier, the integral term is non-negative and equals zero if and only if $T \in \mathcal{R}(W^\epsilon)$. Therefore, any optimal process

must belong to the reciprocal class, reducing the dynamic problem over Ω to the static Entropic Optimal Transport (EOT) problem over the endpoint couplings:

$$\inf_{T\in\mathcal{F}(P_0,P_1)} \mathrm{KL}(T \mid W^\epsilon) = \inf_{\pi\in\Pi(P_0,P_1)} \mathrm{KL}(\pi \mid \pi^{W^\epsilon}), \quad T \in \mathcal{R}(W^\epsilon)$$

The static optimization problem over $\Pi(P_0, P_1)$ corresponds to the Entropic Optimal Transport (EOT) problem. Expanding the KL divergence with respect to the Wiener transition kernel, we have:

$$\mathrm{KL}(\pi \mid \pi^{W^\epsilon}) = \int_{\mathcal{X}\times\mathcal{Y}} \frac{\|x-y\|^2}{2\epsilon} \mathrm{d}\pi(x,y) - \mathrm{H}(\pi) + C,$$

where $\mathrm{H}(\pi)$ denotes the differential entropy of π, and C is a constant depending only on P_0 and ϵ.

Because of the scaling factor 2ϵ and the constant C, minimizing the KL divergence is equivalent to finding the minimizer for the standard EOT problem with a quadratic cost and regularization parameter 2ϵ:

$$\inf_{\pi\in\Pi(P_0,P_1)} \mathrm{KL}(\pi \mid \pi^{W^\epsilon}) = \inf_{\pi\in\Pi(P_0,P_1)} \left\{ \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \mathrm{d}\pi^T(x,y) - 2\epsilon \mathrm{H}(\pi^T) \right\}$$

Let $\mathcal{C}_{b,2}(\mathcal{Y})$ be the space of continuous functions bounded by a quadratic, defined rigorously as:

$$\mathcal{C}_{b,2}(\mathcal{Y}) = \left\{ \beta \in C(\mathcal{Y}, \mathbb{R}) : \exists u, v, w \in \mathbb{R} \text{ such that } u\|y\|^2 + v \le \beta(y) \le w \text{ for all } y \in \mathcal{Y} \right\}.$$

Using the semi-dual formulation of the EOT objective, the minimization can be recast as a maximization over β:

$$\inf_{\pi\in\Pi(P_0,P_1)} \mathrm{KL}(\pi \parallel \pi^{W^\epsilon}) = \sup_{\beta\in\mathcal{C}_{b,2}(\mathcal{Y})} \left\{ \int_{\mathcal{X}} B^\beta(x) \mathrm{d}P_0(x) + \int_{\mathcal{Y}} \beta(y) \mathrm{d}P_1(y) \right\},$$

where the function $B^\beta : \mathcal{X} \to \mathbb{R}$ is defined as:

$$B^\beta(x) = \inf_{\nu\in\mathcal{P}_2(\mathcal{Y})} \left\{ \mathrm{KL}(\nu \mid \pi^{W^\epsilon}(\cdot \mid x)) - \int_{\mathcal{Y}} \beta(y) \mathrm{d}\nu(y) \right\}.$$

Let $\mathcal{D}(P_0) \subset \mathcal{P}(\Omega)$ denote the space of all finite-energy diffusion measures with initial distribution P_0. Specifically, a measure $T_f \in \mathcal{D}(P_0)$ is the law of a continuous-time process $(X_t)_{t\in[0,1]}$ governed by the following stochastic differential equation (SDE):

$$\mathrm{d}X_t = f(t, X_t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t, \quad X_0 \sim P_0$$

where the drift vector field $f : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ satisfies the finite-energy condition:

$$\mathbb{E}_{T_f}\left[\int_0^1 \|f(t, X_t)\|^2 \mathrm{d}t\right] = \int_\Omega \int_0^1 \|f(t, X_t(\omega))\|^2 \mathrm{d}t \, \mathrm{d}T_f(\omega) < \infty.$$

Following the authors of Entropic Neural Optimal Transport via Diffusion Processes, for finite-energy diffusions, we can rewrite the objective functional of the dynamic Schrödinger Problem as:

$$\mathrm{KL}(T \mid W^\epsilon) = \frac{1}{2\epsilon}\, \mathbb{E}_{T_f}\left[\int_0^1 \|f(t, X_t)\|^2 \mathrm{d}t\right]$$

Consequently, the Schrödinger Problem reduces to the following minimization problem:

$$\inf_{T \in \mathcal{F}(P_0, P_1)} \mathrm{KL}(T \mid W^\epsilon) = \inf_{T_f \in \mathcal{D}(P_0, P_1)} \frac{1}{2\epsilon}\, \mathbb{E}_{T_f}\left[\int_0^1 \|f(t, X_t)\|^2 \mathrm{d}t\right]$$

where $\mathcal{D}(P_0, P_1) \subset \mathcal{P}(\Omega)$ denotes the space of all finite-energy diffusion measures on Ω having marginal distributions P_0 and P_1 at times $t = 0$ and $t = 1$, respectively.

Let the Lagrangian functional $\mathcal{L}(\beta, T_f) : \mathcal{C}_{b,2}(\mathcal{Y}) \times \mathcal{D}(P_0) \to \mathbb{R}$ be defined as:

$$\mathcal{L}(\beta, T_f) = \mathbb{E}_{T_f}\left[\frac{1}{2\epsilon}\int_0^1 \|f(t, X_t)\|^2 \mathrm{d}t\right] - \int_{\mathcal{Y}} \beta(t) \mathrm{d}\pi^{T_f}(y) + \int_{\mathcal{Y}} \beta(y) \mathrm{d}P_1(y)$$

where $\pi_1^{T_f}$ is the marginal distribution of the process T_f at time $t = 1$.

Theorem 1. Consider the following saddle point optimization problem:

$$\sup_{\beta \in \mathcal{C}_{b,2}(\mathcal{Y})} \inf_{T_f \in \mathcal{D}(P_0)} \mathcal{L}(\beta, T_f)$$

If (β^*, T_f^*) is an optimal pair for this minimax problem, then T_f^* is the solution to the dynamic Schrödinger Problem.

Furthermore, define the scaled Lagrangian functional $\tilde{\mathcal{L}}(\tilde{\beta}, T_f)$ as:

$$\tilde{\mathcal{L}}(\tilde{\beta}, T_f) = \mathbb{E}_{T_f}\left[\int_0^1 \|f(t, X_t)\|^2 \mathrm{d}t\right] - \int_{\mathcal{Y}} \tilde{\beta}(t) \mathrm{d}\pi^{T_f}(y) + \int_{\mathcal{Y}} \tilde{\beta}(y) \mathrm{d}P_1(y)$$

Notice the algebraic relationship between the original and scaled functionals. By substituting $\tilde{\beta} = 2\epsilon\beta$, we can factor out the scalar:

$$\tilde{\mathcal{L}}(2\epsilon\beta, T_f) = 2\epsilon \mathcal{L}(\beta, T_f).$$

Since $\epsilon > 0$ and the function space $\mathcal{C}_{b,2}(\mathcal{Y})$ is closed under scalar multiplication (meaning $\beta \in \mathcal{C}_{b,2}(\mathcal{Y})$ if and only if $2\epsilon\beta \in \mathcal{C}_{b,2}(\mathcal{Y})$), scaling the objective by a positive constant does not change the optimal diffusion measure T_f^* or the effective dual constraints.

Therefore, we can absorb the factor of 2ϵ into the dual variable. Dropping the tilde for notational convenience, we arrive at the following equivalent saddle-point optimization problem:

$$\sup_{\beta \in \mathcal{C}_{b,2}(\mathcal{Y})} \inf_{T_f \in \mathcal{D}(P_0)} \left\{ \mathbb{E}_{T_f} \left[\int_0^1 \|f(t, X_t)\|^2 \mathrm{d}t \right] - \int_{\mathcal{Y}} \beta(y) \mathrm{d}\pi_1^{T_f}(y) + \int_{\mathcal{Y}} \beta(y) \mathrm{d}P_1(y) \right\}.$$

To solve the optimization problem, we parameterize the drift function $f(t, x)$ of the diffusion process and the dual potential $\beta(y)$ using neural networks:

$$f_\theta : [0, 1] \times \mathbb{R}^d \to \mathbb{R}^d, \qquad \beta_\phi : \mathbb{R}^d \to \mathbb{R}.$$

Evaluating the objective functional requires sampling trajectories from the parameterized diffusion measure T_{f_θ}, which is governed by the SDE:

$$\mathrm{d}X_t = f_\theta(t, X_t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t, \quad X_0 \sim P_0.$$

We simulate this process using the Euler-Maruyama discretization scheme over N uniform time steps of size $\Delta t = 1/N$. Letting $t_n = n\Delta t$, the discrete-time transitions are:

$$X_{n+1} = X_n + f_\theta(t_n, X_n)\Delta t + \sqrt{\epsilon \Delta t} Z_n, \quad Z_n \sim N(0, \mathbf{I}).$$

By simulating M independent trajectories, we generate a batch of terminal states $\{X_N^{(m)}\}_{m=1}^M$, which serve as empirical samples from the time $t = 1$ marginal distribution $\pi_1^{T_{f_\theta}}$.

The time integral inside the expectation is approximated using a left-point Riemann sum over the discretization grid:

$$\int_0^1 \|f_\theta(t, X_t)\|^2 \mathrm{d}t \approx \Delta t \sum_{n=0}^{N-1} \|f_\theta(t_n, X_n)\|^2.$$

Combining this with a Monte Carlo approximation over the M simulated trajectories, we estimate the energy term (which we denote as $\widehat{\mathcal{E}}(\theta)$):

$$\mathbb{E}_{T_{f_\theta}} \left[\int_0^1 \|f_\theta(t, X_t)\|^2 \mathrm{d}t \right] \approx \frac{\Delta t}{M} \sum_{m=1}^M \sum_{n=0}^{N-1} \|f_\theta(t_n, X_n^{(m)})\|^2 \equiv \widehat{\mathcal{E}}(\theta).$$

By drawing a batch of target samples $\{y^{(m)}\}_{m=1}^M \sim P_1$ alongside our simulated endpoints $\{X_N^{(m)}\}_{m=1}^M$, we formulate the empirical min-max objective function:

$$\hat{J}(\phi, \theta) = \widehat{\mathcal{E}}(\theta) - \frac{1}{M} \sum_{m=1}^M \beta_\phi(X_N^{(m)}) + \frac{1}{M} \sum_{m=1}^M \beta_\phi(y^{(m)}).$$

To optimize this saddle-point problem, we compute the gradients with respect to the network parameters ϕ and θ. The gradient for the dual potential (ascent) is:

$$\nabla_\phi \hat{J}(\phi, \theta) = -\frac{1}{M}\sum_{m=1}^{M} \nabla_\phi \beta_\phi(X_N^{(m)}) + \frac{1}{M}\sum_{m=1}^{M} \nabla_\phi \beta_\phi(y^{(m)}).$$

The gradient for the drift network (descent) requires differentiating through the SDE rollout (e.g., via backpropagation through time), yielding:

$$\nabla_\theta \hat{J}(\phi, \theta) = \nabla_\theta \widehat{\mathcal{E}}(\theta) - \frac{1}{M}\sum_{m=1}^{M} \nabla_\theta \beta_\phi(X_N^{(m)}).$$

Algorithm 1: Entropic Neural OT (ENOT)

Input: samples from P_0, P_1, noice variance $\epsilon > 0$, drift network f_θ, beta network β_ϕ, number of discretization steps N, number of inner iterations K_f.

while *Not Converged* **do**

Sample batches $X_0 \sim P_0$, $Y \sim P_1$

$\{X_n, f_n\}_{n=0}^{N-1} \leftarrow \text{Eul_Mar}(X_0, f_\theta)$

$$\mathcal{L}_\beta \leftarrow \frac{1}{|X_N|}\sum_{x \in X_N} \beta_\phi(x) - \frac{1}{|Y|}\sum_{y \in Y} \beta_\phi(y)$$

Update parameters ϕ using $\partial_\phi \mathcal{L}_\beta$

for $k = 1$ **to** K_f **do**

Sample batches $X_0 \sim P_0$, $Y \sim P_1$

$\{X_n, f_n\}_{n=0}^{N-1} \leftarrow \text{Eul_Mar}(X_0, f_\theta)$

$$\widehat{\mathcal{E}}(\theta) \leftarrow \frac{1}{N}\sum_{n=0}^{N-1} \frac{1}{M}\sum_{m=1}^{M} \|f_n^{(m)}\|^2$$

$$\mathcal{L}_f \leftarrow \widehat{\mathcal{E}}(\theta) - \frac{1}{|X_N|}\sum_{x \in X_N} \beta_\phi(x)$$

Update parameters θ using $\partial_\theta \mathcal{L}_f$

Output: Drift f_θ^* of the Schrodinger Bridge $T_{f_\theta^*}$.

Algorithm 2: Euler-Maruyama

Input: Batch of initial states X_0 at time $t = 0$, drift netwrok f_θ, number of steps N, noise variance $\epsilon \geq 0$.

$\Delta t = 1/N$
for $t = 1$ **to** N **do**
  for $i = 1$ **to** $|X_0|$ **do**
    Sample noise $W \sim N(0, \mathbf{I})$

$$f_{t-1,i} \leftarrow f(t-1, X_{t-1,i})$$
$$X_{t,i} \leftarrow X_{t-1,i} + f_{t-1,i}\Delta t + \sqrt{\epsilon \Delta t} W$$

Output: Batches $\{X_n\}_{n=1}^N$
of intermediate steps at $t = n/N$ simulating the process
$\mathrm{d}X_t = f_\theta(t, X_t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t$, batches $\{f_n\}_{n=1}^N$ simulating the process $f(t_n, X_n)$.

The methodology detailed above is not restricted to a scaled Brownian motion reference. It naturally extends to reference processes with a non-zero drift, such as the Ornstein-Uhlenbeck (OU) process. Let $Q_u \in \mathcal{P}(\Omega)$ denote the law of a reference diffusion process governed by the following SDE:

$$\mathrm{d}X_t = u(t, X_t)\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}W_t, \quad X_0 \sim P_0,$$

where $u : [0, 1] \times \mathbb{R}^d \to \mathbb{R}^d$ is a predefined reference drift vector field.

By applying Girsanov's Theorem relative to this new reference measure, the energy functional shifts to penalize the difference between the learned drift and the reference drift. The dynamic Schrödinger Problem thus becomes:

$$\inf_{T_f \in \mathcal{D}(P_0, P_1)} \mathrm{KL}(T_f \parallel Q_u) = \inf_{T_f \in \mathcal{D}(P_0, P_1)} \frac{1}{2\epsilon} \mathbb{E}_{T_f} \left[\int_0^1 \|f(t, X_t) - u(t, X_t)\|^2 \mathrm{d}t \right].$$

By following the exact same semi-dual formulation and scaling argument used for the Wiener reference, we can absorb the 2ϵ factor into the dual potential β. The modified Lagrangian functional incorporates the shifted energy term, yielding the updated saddle-point optimization problem:

$$\sup_{\beta \in \mathcal{C}_{b,2}(\mathcal{Y})} \inf_{T_f \in \mathcal{D}(P_0)} \left\{ \mathbb{E}_{T_f} \left[\int_0^1 \|f(t, X_t) - u(t, X_t)\|^2 \mathrm{d}t \right] - \int_{\mathcal{Y}} \beta(y) \mathrm{d}\pi_1^{T_f}(y) + \int_{\mathcal{Y}} \beta(y) \mathrm{d}P_1(y) \right\}.$$